

PHYSICS STRATEGIC ROADMAP

2026

Comprehensive Topic Analysis Based on Four Targeted Board Paper Sets

1. RAY OPTICS & OPTICAL INSTRUMENTS (8-10 MARKS)

Target Areas: Compound Microscope & Astronomical Telescope (Adjustment & Magnification), Lens Maker's Formula, Prism (Refractive Index vs. Deviation).

Recurring Pro-Tip: Always draw Ray Diagrams with arrows. Missing arrows often leads to a 1-mark deduction in 5-mark questions.

2. SEMICONDUCTOR ELECTRONICS (7 MARKS)

- **Biasing:** Impact of Forward/Reverse bias on Depletion Layer thickness.
- **Rectification:** Working of Full-wave Rectifier with Input/Output waveforms.
- **Semiconductors:** Band gap diagrams distinguishing n-type and p-type.

3. WAVE OPTICS (6-7 MARKS)

YDSE Fringe Width

Single Slit Diffraction

Huygens Principle (Reflection/Refraction)

Common Trap: Differentiating between "interference" and "diffraction" pattern properties (e.g., central maximum width).

4. CURRENT ELECTRICITY & MOVING CHARGES

Unit	High-Yield Topic	Frequency
Current	Drift velocity relationship with Temperature	Very High
Magnetism	Galvanometer to Voltmeter Conversion	High
Dipole	Torque on a loop in Magnetic Field	Medium

5. MODERN PHYSICS (ATOMS/NUCLEI/DUAL NATURE)

- **Photoelectric Effect:** Stopping Potential vs Frequency graphs (Threshold frequency/ Work function).
- **Bohr's Orbit:** Radius and Energy derivations (specifically $n=1, 2, 3$ values).
- **Binding Energy:** Calculation of Mass Defect; standard 1:1 Nuclear Density ratio for Nuclei.

6. QUICK FORMULAS & DIAGRAMS TO MASTER

- **Capacitors:** Energy stored = $\frac{1}{2} CV^2$; Effect of Dielectric (K) on V , Q , C , and E .
- **LCR Circuit:** Resonance condition ($X_L = X_C$) and Power Factor.
- **Transformers:** Principle of Mutual Induction; Eddy current & Flux leakage losses.